GWAS for serum levels of the liver enzyme AST

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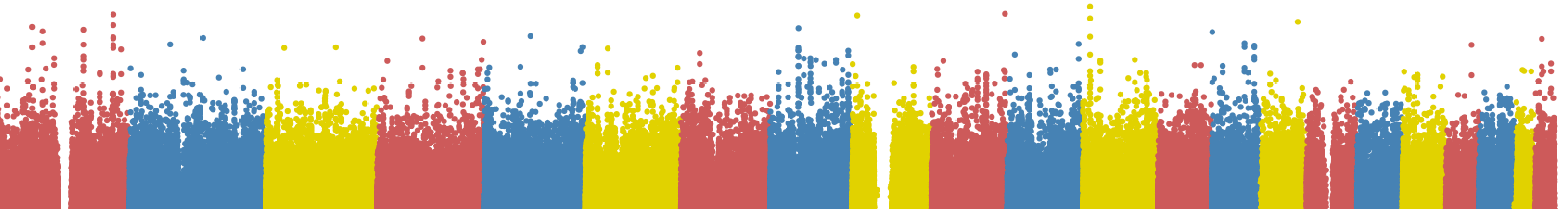


Aspartate Aminotransferase (AST)

Biochemistry
A key enzyme in amino acid metabolism; it catalyzes the reversible transfer of an α-amino group between aspartate and α-ketoglutarate to form oxaloacetate and glutamate.

Distribution & Expression
It has two isoenzymes (cytosolic and mitochondrial). It is present in many tissues

BIomarker
Under normal conditions, plasma AST levels are low. During cell lysis, apoptosis, or damage to the integrity of the hepatocyte membrane, the enzyme is released into the bloodstream, serving as a highly sensitive biomarker of liver damage

(Ndrepepa, 2021)

PennCath Cohort and Omic data

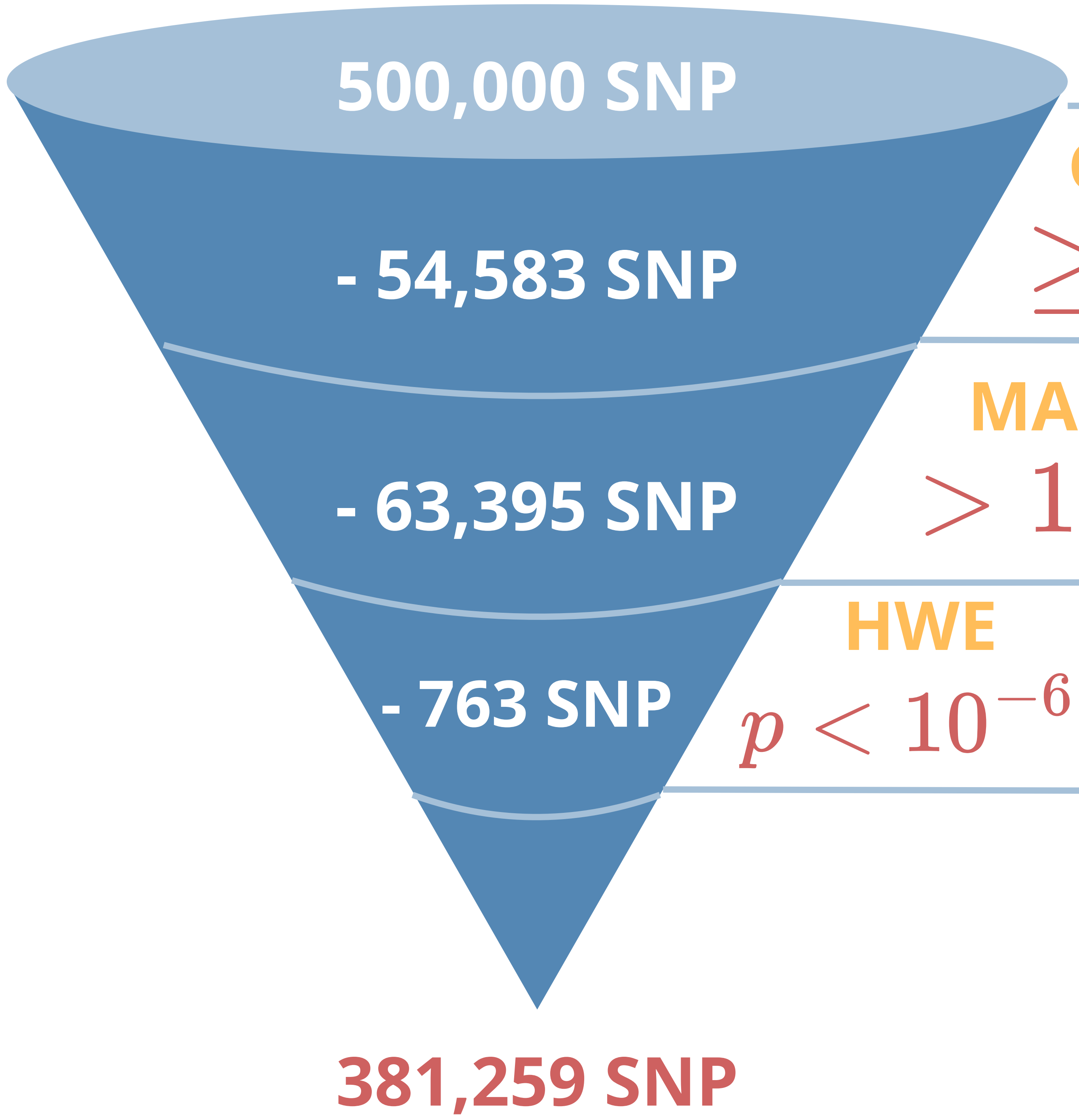
Study Design

Hospital-based cohort using a case-control design for coronary artery disease (CAD).

Category	Component / Files	Details and Initial Dimensions
Genomic Data	PLINK files (.bed, .bim, .fam)	1,401 initial individuals and 500,000 SNPs.
	binary genotype matrix (.bed)	Allelic dose coding (0, 1, 2) for additive models.
Clinical Data	phenotype file (Clinical_AST.csv)	Quantitative outcome variable: Serum AST (U/L).
	Associated covariates	Age, Sex, BMI, Alcohol consumption and Smoking status.
Structural Design	CAD status (from GWAS_clinical.csv)	Variable added to preserve and control for the cohort's original methodological design.

Quality Control

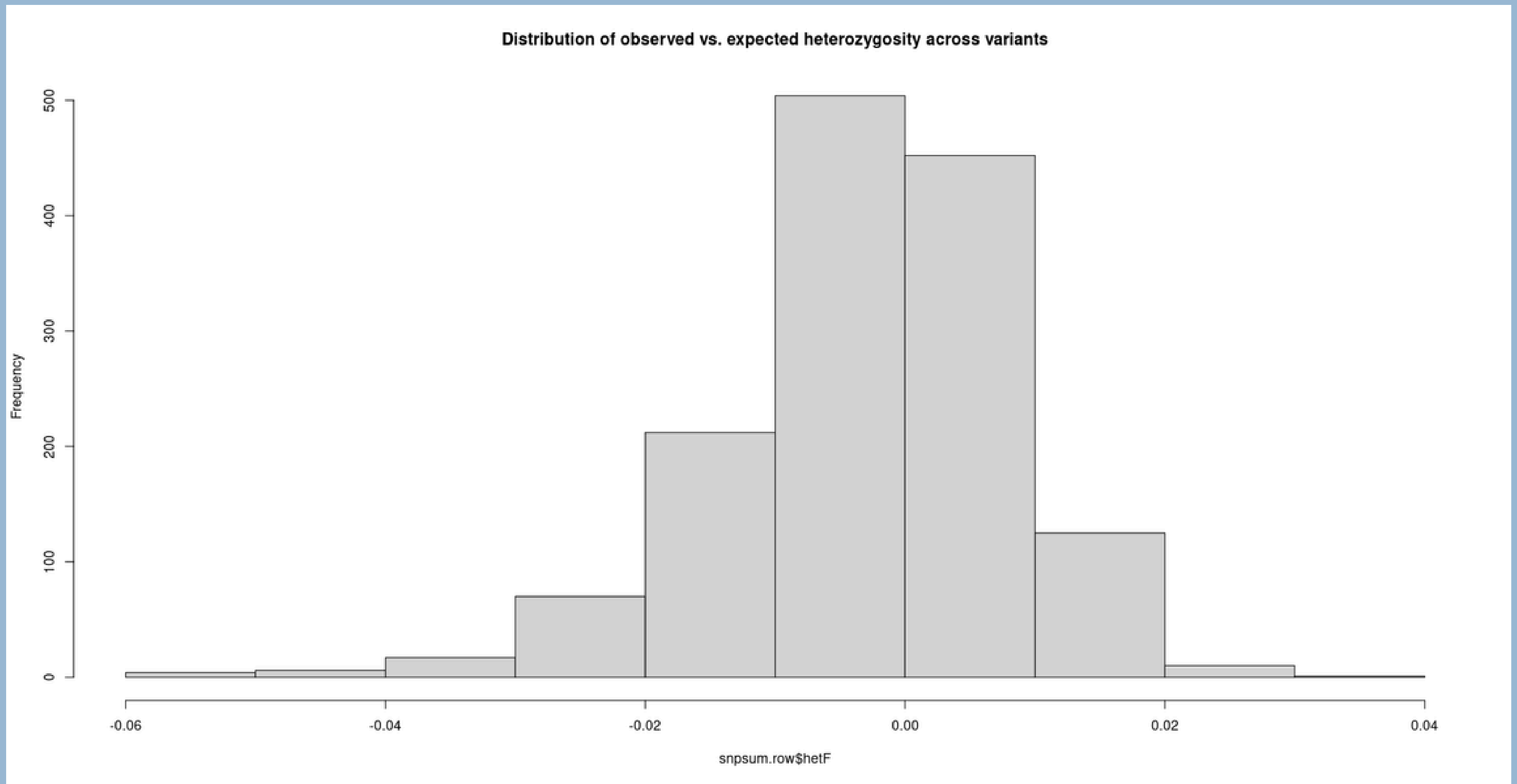
Variant Filter (SNPs)

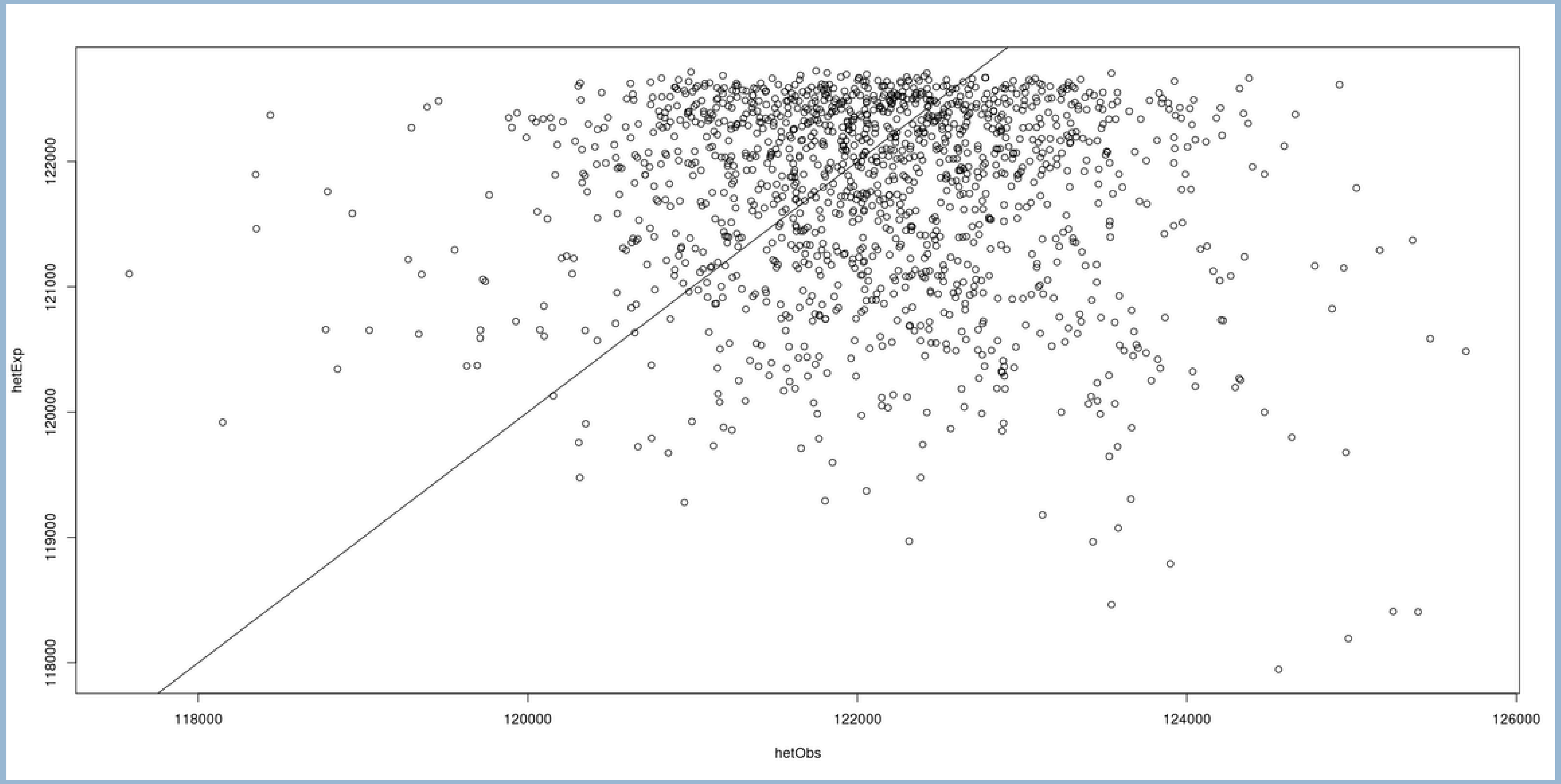


Call rate $\geq 95\%$ — Elimination of technical hybridization errors in the microarray.

MAF $> 1\%$ — Exclusion of rare variants due to lack of statistical power.

HWE $p < 10^{-6}$ — Detection of systematic genotyping errors





Call Rate per Individual $\geq 97\%$

Evaluates the quality and integrity of the DNA extracted from each patient.

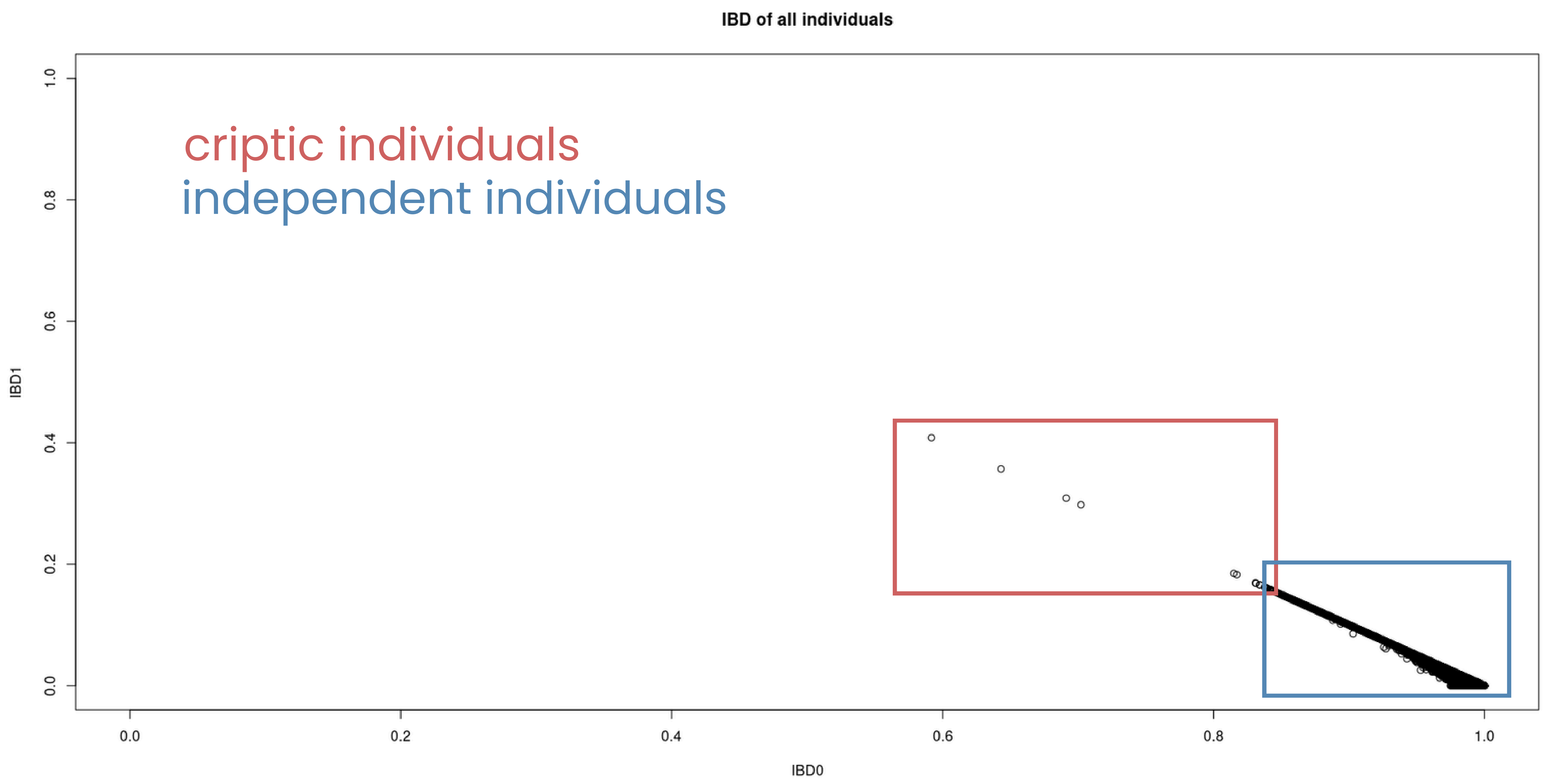
1 individual was excluded due a high rate of missing data across the entire microarry

Heterozygosity Index $|F| \leq 0.05$

Compare the proportion of heterozygous genotypes observed with the expected proportion based on probability (HWE).

4 individual were excluded here

Criptic Relatedness (IBD)



LD pruning $r^2 \geq 0.2$

Removal of variants to retain only statistically independent SNPs.

IBD measure

Pairwise comparison estimating the probability of sharing zero (IBD0) or one allele (IBD1) from a recent common ancestor.

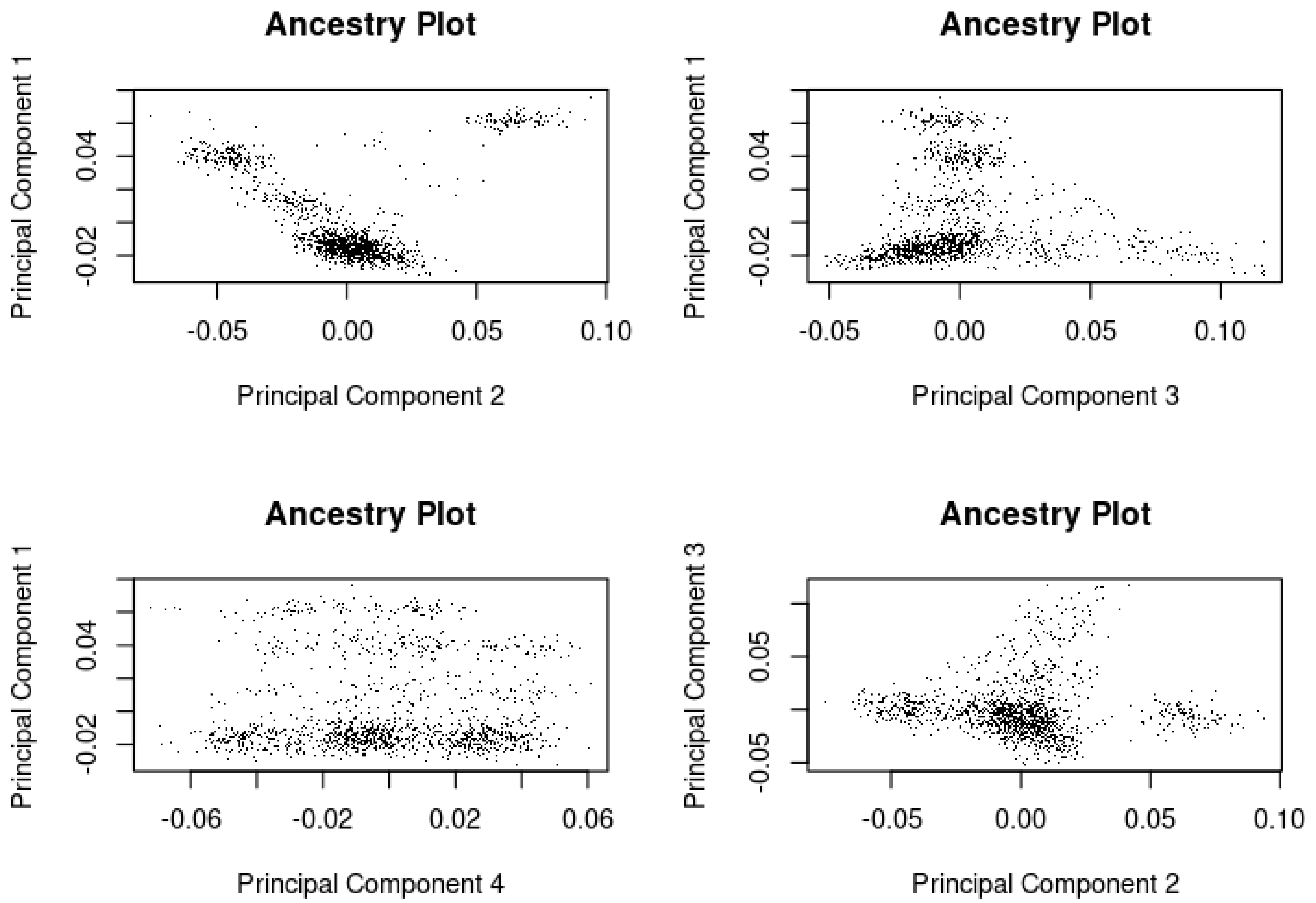
15 individuals were excluded (one member at random from each pair) to ensure the independence of the observations.

Ancestry (PCA)

Population Stratification

Differences in allele frequencies due to ancestral history can create spurious associations with the phenotype.

With PCA we can identify the axes with the greatest genetic variation using independent SNPs (LD-pruned)



The first **4 PC** were extracted to be included as covariates in the regression model.

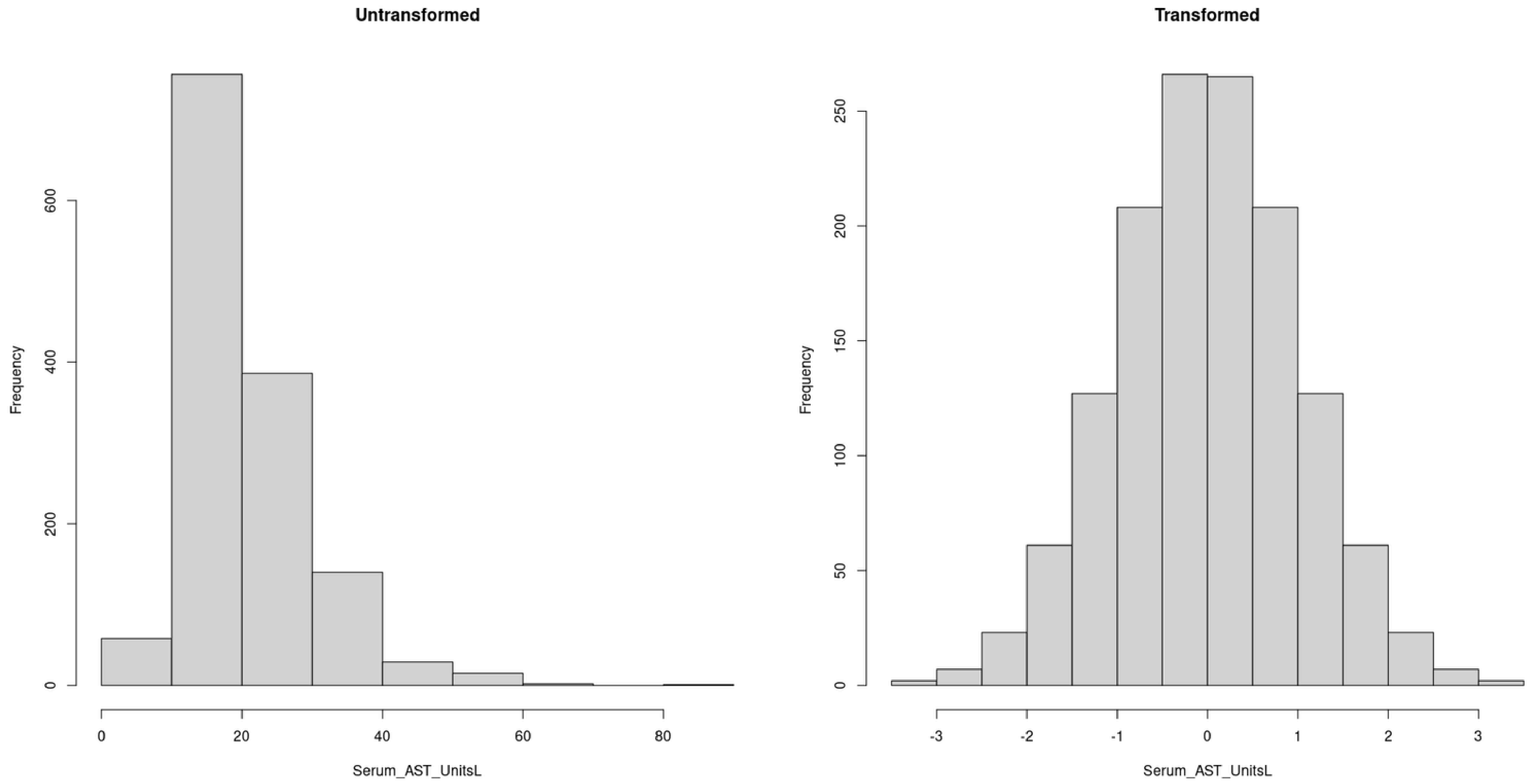
GWAS statistical model

381,255 SNPs are included in the GWAS
1,388 Individual are included in the GWAS

Phenodata & Transformation

- An **inverse rank normal transformation** (rntransform) was applied to the outcome variable

- To mitigate the violation of the **normality assumption required by linear regression.**



To **generate the phenodata**, clinical data (covariates and transformed outcomes) were merged with the genetic coordinates of the first 4 PC.

Linear Regression Model

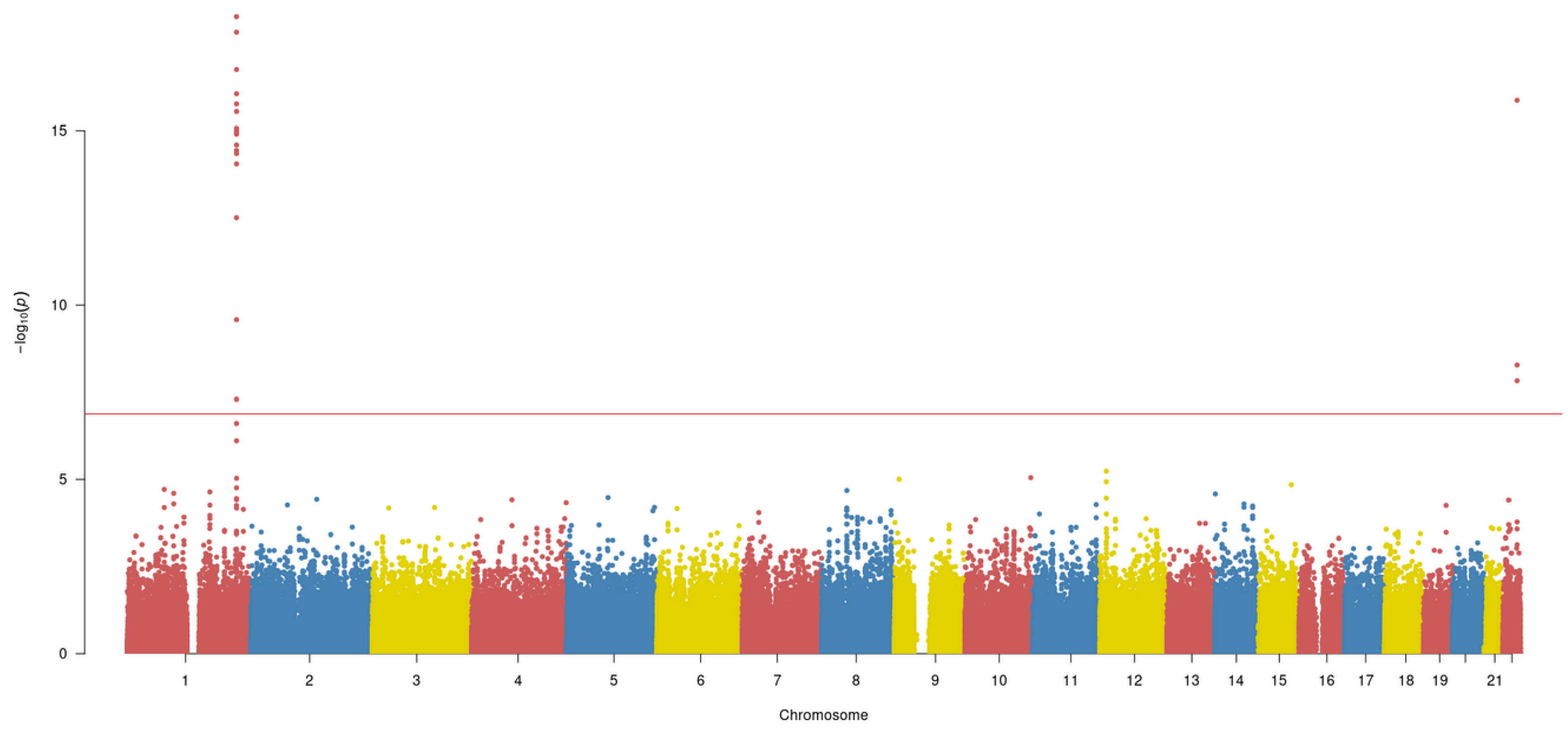
$$Y_{RINT(AST)} = \beta_0 + \beta_1(Sex) + \beta_2(Age) + \beta_3(Alcohol) + \beta_4(BMI) + \sum_{i=1}^{4} \gamma_i(PC_i) + \epsilon$$

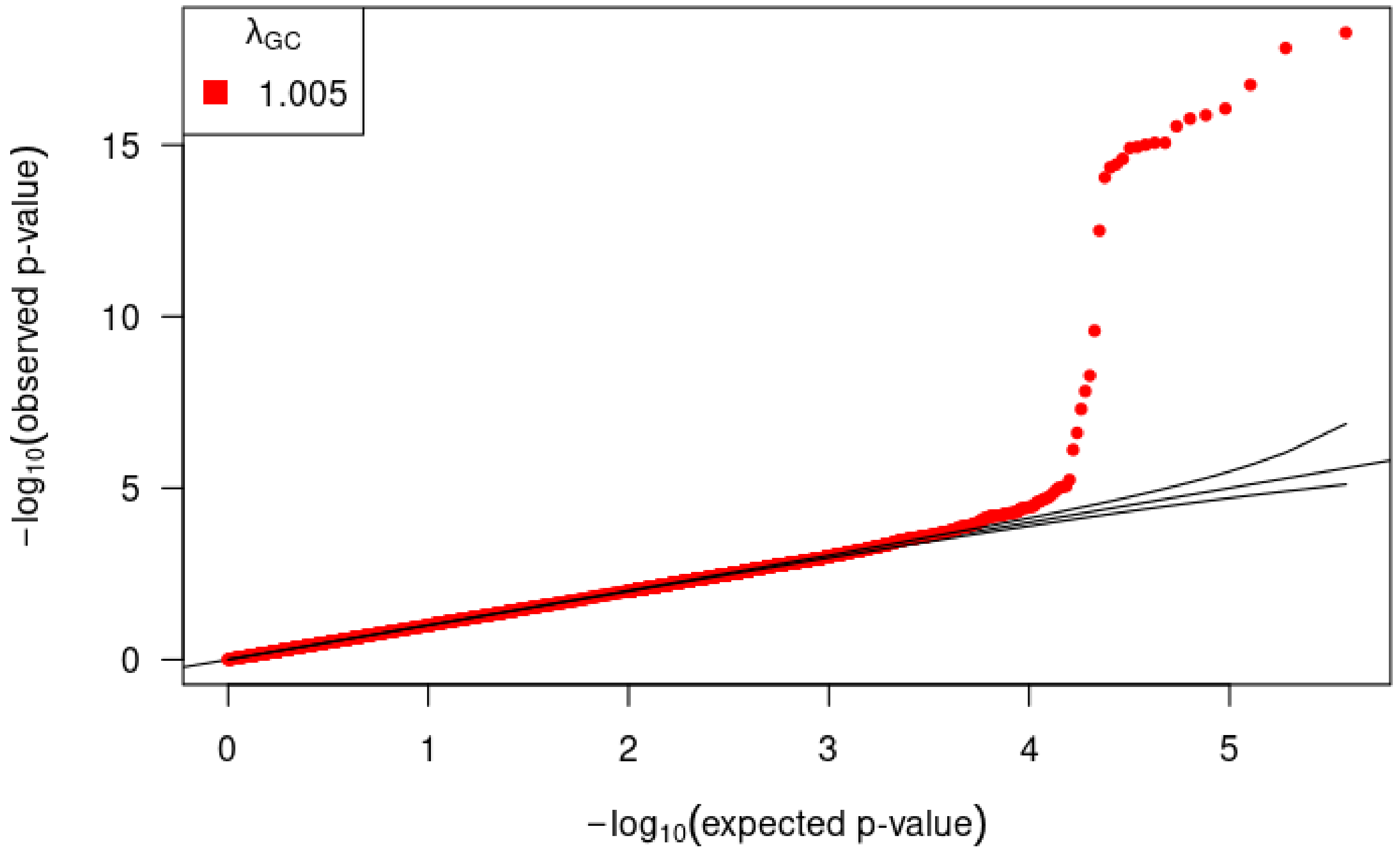
Where Y is the transformed phenotype, β_0 is the intercept, the β are the clinical effect estimators, the γ are the ancestry estimators (PCA), and ϵ is the residual error.

Covariate	Estimate (β)	Std. Error	p-value
Intercept	-1.8	0.273	5.75×10^{-11}
Sex (Female vs Male)	-0.142	0.058	0.014
Age (Years)	0.012	0.002	3.67×10^{-5}
Alcohol (Drinks/wk)	0.069	0.015	5.33×10^{-6}
BMI	0.034	0.007	4.37×10^{-6}
PC1 (Ancestry)	4.925	0.979	5.59×10^{-7}
PC2 (Ancestry)	-2.864	0.975	0.003
PC3 & PC4	-	-	> 0.53

Global result of GWAS

Computed 342,575 independent linear regression models by iterating over genotypes, adjusted for clinical covariates and ancestry factors.





A massive and highly significant signal located on chromosome 1, far exceeding the threshold for genomic significance
(Bonferroni correction: $p < 1.31 \times 10^{-7}$)

It demonstrates a perfect alignment between the distribution of observed statistics and the expected value

Variant Top Hits & Annotation

SNP (rsID)	Chr	Posición Física (BP)	A1 / A2	Estimate (β)	P-value	Location	Gene Symbol
rs2275707	**1**	**220,088,047**	**G / T**	**0.434**	$\mathbf{5.34 \times 10^{-19}}$	**threeUTR/ intron**	**SLC30A10**
rs1776046	1	220,110,560	G / A	0.403	1.49×10^{-18}	intron	RABGAP1L-DT
rs2647438	1	220,152,251	C / T	0.412	1.77×10^{-17}	NA	NA
rs2577138	1	220,205,980	A / C	0.385	8.77×10^{-17}	intron	TTYTY1B
rs734561	22	44,324,104	A / G	0.364	1.35×10^{-16}	intron	QPCTL

The top 4 SNPs map to the same genomic window of approximately **117 kb on Chromosome 1.**

UCSC

Curated Genomic Annotation

SNP (rsID)	Chr	Posición Física (BP)	Location	Gene Symbol (hg19)	Gene Symbol (GRCh18)
rs2275707	1	220,088,047	**threeUTR/ intron**	**SLC30A10**	**SLC30A10**
rs1776046	1	220,110,560	intron	RABGAP1L-DT	SLC30A10
rs2647438	1	220,152,251	intron	NA	EPRS1
rs2577138	1	220,205,980	intron	TTYTY1B	EPRS1
rs734561	22	44,324,104	intron	QPCTL	PNPLA3

The only **variant reported** as having **clinical significance** (ClinVar) is rs2275707

NCBI

rs2275707

ClinVar
Clinically relevant variation

NM_018713.3(SLC30A10):c.*744G>T

Gene:	SLC30A10:solute carrier family 30 member 10 [Gene - OMIM - HGNC]
Variant type:	single nucleotide variant
Cytogenetic location:	1q41
Genomic location:	Chr1: 219914705 (on Assembly GRCh38) Chr1: 220088047 (on Assembly GRCh37)
Preferred name:	NM_018713.3(SLC30A10):c.*744G>T
HGVS:	NC_000001.11:g.219914705C>A NG_032153.2:g.18947G>T NM_001376929.1:c.*744G>T ...more
Links:	dbSNP: rs2275707
Molecular consequence:	NM_001376929.1:c.*744G>T - 3 prime UTR variant - [Sequence Ontology: SO:0001624] NM_018713.3:c.*744G>T - 3 prime UTR variant - [Sequence Ontology: SO:0001624] NR_046437.2:n.2357G>T - non-coding transcript variant - [Sequence Ontology: SO:0001619] NR_165031.1:n.2094G>T - non-coding transcript variant - [Sequence Ontology: SO:0001619]

Condition(s)

Name:	Hypermanganesemia with dystonia, polycythemia, and cirrhosis (HMNDYT1)
Synonyms:	Hepatic Cirrhosis, Dystonia, Polycythemia and Hypermanganesemia; Cirrhosis - dystonia - polycythemia - hypermanganesemia syndrome; Hypermanganesemia with Dystonia 1
Identifiers:	MONDO: MONDO:0013208; MedGen: C2750442; Orphanet: 309854; OMIM: 613280

https://www.ncbi.nlm.nih.gov/clinvar/RCV000364944.5/

OMIM
Human Genetics Knowledge for the World

613280

HYPERMANGANESEMIA WITH DYSTONIA 1; HMNDYT1

Alternative titles; symbols

HYPERMANGANESEMIA WITH DYSTONIA, POLYCYTHEMIA, AND CIRRHOSIS; HMDPC

Phenotype-Gene Relationships

Location	Phenotype	Phenotype MIM number	Inheritance	Phenotype mapping key	Gene/Locus	Gene/Locus MIM number
1q41	Hypermanganesemia with dystonia 1	613280	AR	3	SLC30A10	611146

Clinical Synopsis | Phenotypic Series | PheneGene Graphics

▼ TEXT

A number sign (#) is used with this entry because of evidence that hypermanganesemia with dystonia-1 (HMNDYT1) is caused by homozygous mutation in the SLC30A10 gene (611146) on chromosome 1q41.

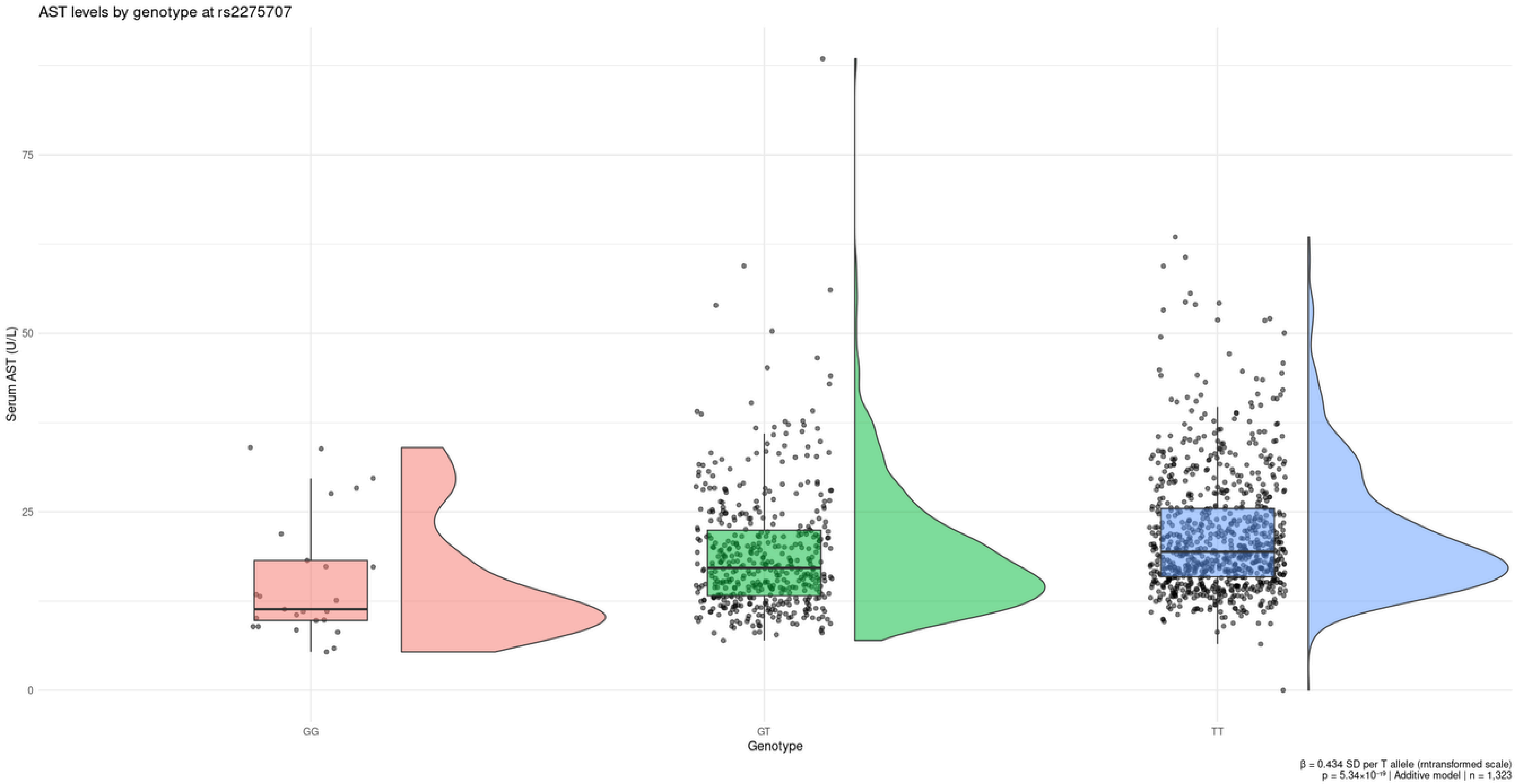
▼ Description

Hypermanganesemia with dystonia-1 (HMNDYT1) is an autosomal recessive metabolic disorder characterized by increased serum manganese, motor neurodegeneration with extrapyramidal features, polycythemia, and hepatic dysfunction, which leads to cirrhosis in some cases. Intellectual function is preserved (summary by Tuschl et al., 2012 and Quadri et al., 2012).

Genetic Heterogeneity of Hypermanganesemia With Dystonia

See also HMNDYT2 (617013), caused by mutation in the SLC39A14 gene (608736) on chromosome 8p21.

https://www.omim.org/entry/613280



Effect size and Genetic Dose

Validation of the Additive Model: A clear dose-dependent stepwise pattern ("additive ladder") is observed.

Estimated effect: $\beta = +0.434$ Standard deviations for each additional risk allele.

SLC30A10, manganese & liver damage

Function of the Gene (SLC30A10):
Encodes a hepatic transporter that is essential for the excretion of manganese (Mn) into bile.

Trade-Off?:
T Allele (Major): Optimal transport function. The liver efficiently filters manganese (Mn), protecting the brain, but the high metabolic flux leads to chronic liver stress (↑ AST).

Allele G (Minority): Reduced expression (-25%). The liver "rests" (↓ baseline AST), but export to bile decreases, raising blood Mn levels and increasing the risk of subclinical neurotoxicity.

(Wahlberg et al., 2016; Ward et al., 2021)

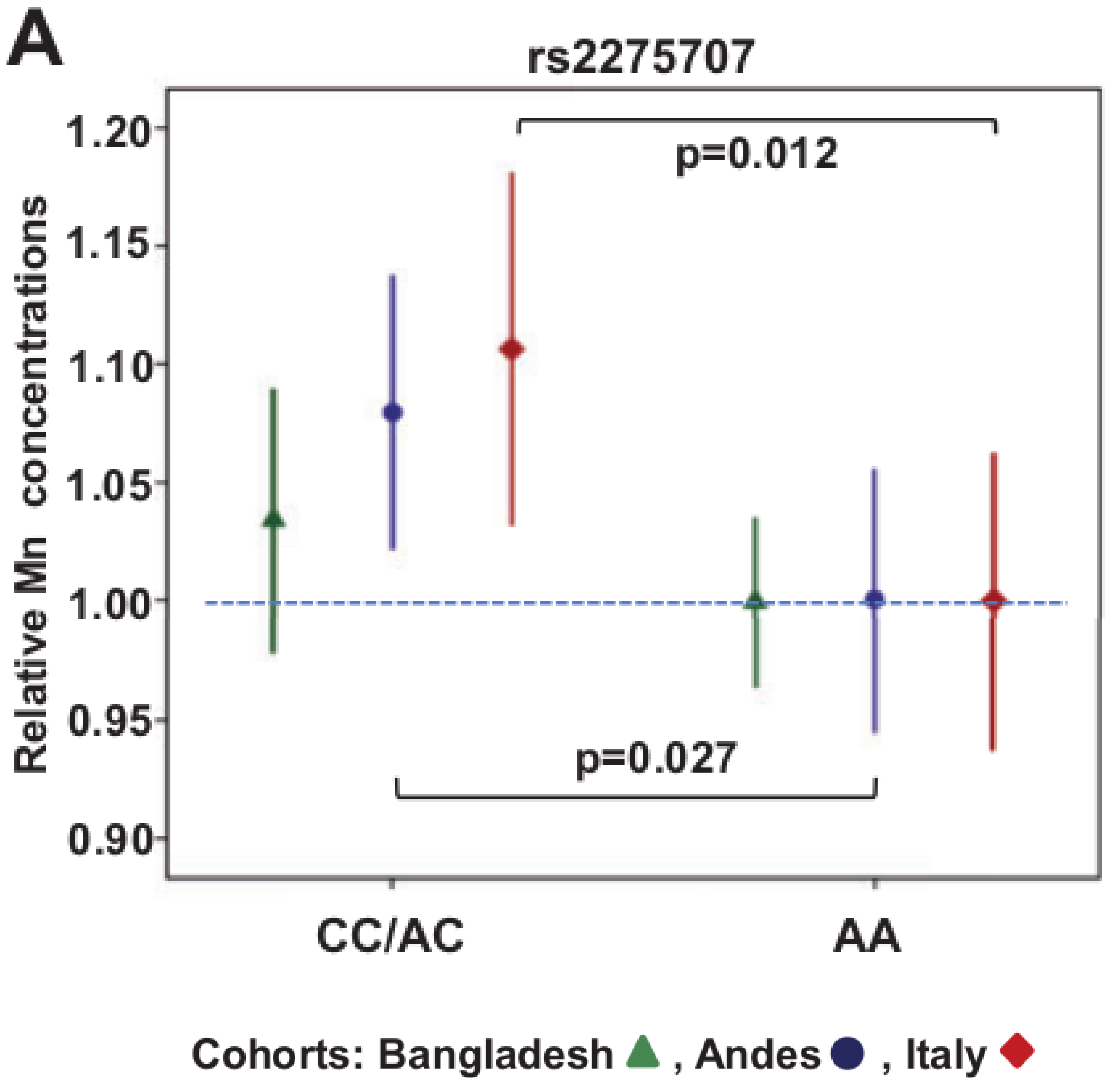


Fig. 2. Associations of genotypes with Mn and SLC30A10 expression. Error bars represent 95% confidence intervals (CI). Due to the low subject number for the rs2275707 variant homozygote genotype (<5%), this was combined with the heterozygote genotype. (A) Associations of SLC30A10 genotypes with blood Mn concentrations (represented by Ery-Mn in Bangladesh and whole blood Mn in the Andes and Italy). To enable comparison between populations, Mn levels are presented relativeto levels for the homozygote genotype of the common allele, which was set to 1. The dashed line marks the reference level.

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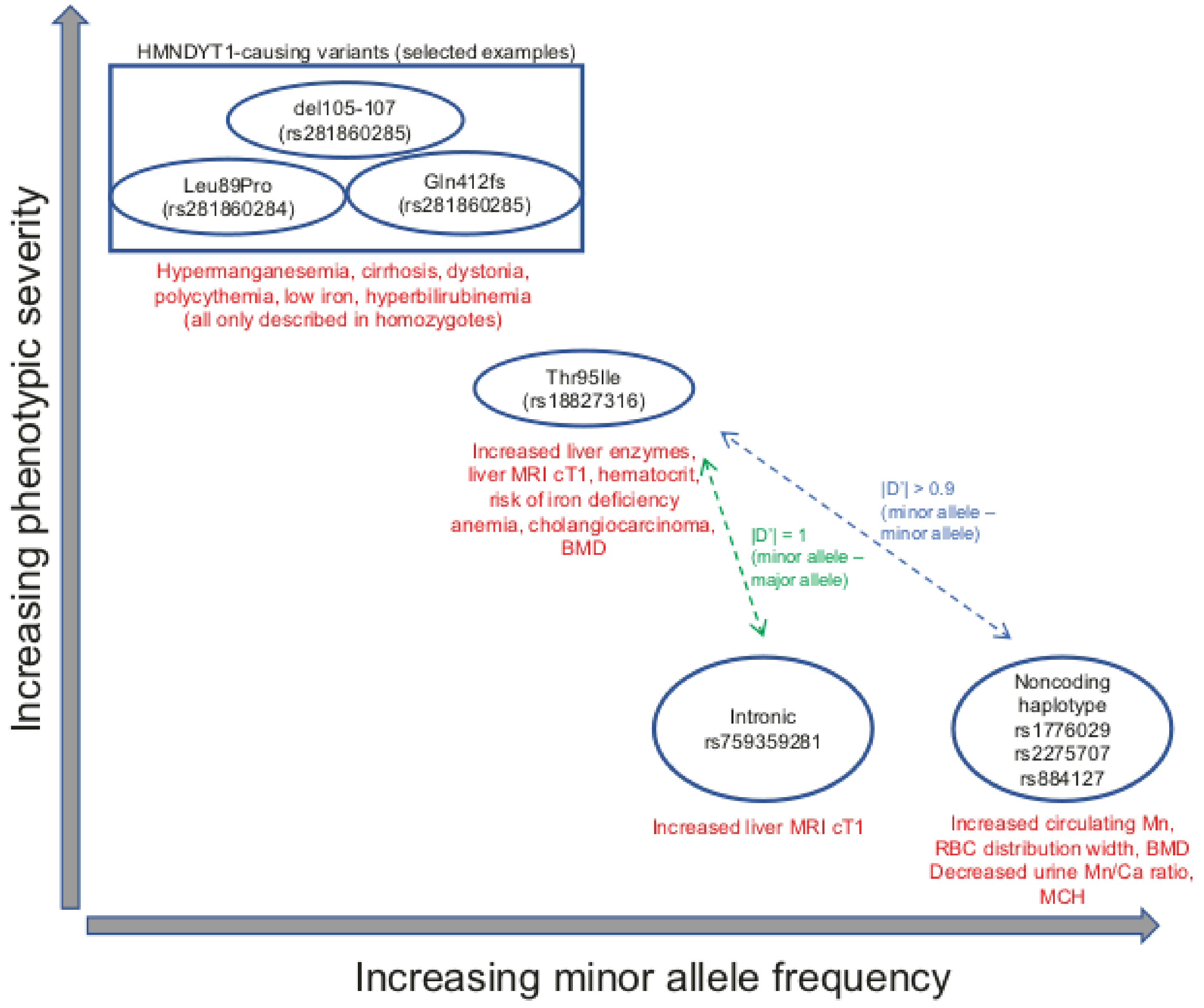

Fig. 7 Relationship of Mendelian and common-variant GWAS phenotypes at the SLC30A10 locus. Phenotypes are summarized for HMNDYT1-causing variants, SLC30A10 Thr95Ile, and GWAS variants.

Take-Home Messages

Filtering (QC) and the additive model (RINT + PCA) completely eliminated population stratification.

The discovery of rs2275707 reveals the pathological spectrum of SLC30A10 the causitive gene for HMNDYT1.

The GWAS detected a strong signal on chromosome 1 in the SLC30A10 gene, revealing a possible evolutionary trade-off between manganese concentration and baseline liver stress.

References

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